# When Positive-Feedback Strategies Crowd: A Family-Level Threshold Framework via Agent-Based Simulation*

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**Abstract**

Volatility targeting (VT), trend-following (TF), and short-horizon mean-reversion (MR) share a common procyclical structure: each strategy's position update is a non-trivial function of recent prices or volatility, so widespread adoption can in principle generate self-reinforcing trading flows. Whether the resulting crowding threshold is VT-specific or a generic property of positive-feedback strategies is unsettled. We build an agent-based model with 1,000 heterogeneous agents, a Kyle (1985) market maker, endogenous VIX dynamics, and 200 fixed noise traders, and use it to test the threshold structure across three strategy classes under matched microstructure. Across 43,300 Monte Carlo simulations spanning 7 adoption levels $\times$ 4 treatments $\times$ 12 strategy-spec cells $\times$ 5 $(\lambda, \gamma)$ OAT cells, three findings emerge. First, under a Sharpe-only detector calibrated to the standalone VT benchmark, VT crosses at 70% adoption—exactly the threshold reported by the canonical analysis. Second, TF and MR cross *at or below* the VT threshold in 12/12 strategy-spec cells and 5/5 microstructure cells, establishing positive-feedback crowding as

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a *family-level* phenomenon. Third, the qualitative ordering survives ±50% perturbations of Kyle-$\lambda$ and VIX-feedback $\gamma$; only the VT magnitude shifts directionally with $\lambda$, consistent with the Kyle-impact mechanism rather than a knife-edge artifact. VT remains the empirically dominant case (largest assets-under-management, real-world deployment), but the threshold is generic.

**Keywords:** volatility targeting, positive feedback, crowding, agent-based model, tipping point, systemic risk

**JEL Classification:** G11, G12, G17, G18

# 1 Introduction

Volatility targeting (VT) has become a ubiquitous tool in dynamic portfolio management. The canonical framework, formalized by Moreira and Muir [2017], prescribes scaling equity exposure as $w_t = \sigma^*/\hat{\sigma}_t$, where $\sigma^*$ is the target volatility and $\hat{\sigma}_t$ is a conditional volatility estimate. Harvey et al. [2018] demonstrate that such strategies meaningfully reduce tail risk across asset classes, and the approach has been widely adopted by pension funds, insurance companies, and target-date funds. Industry estimates suggest that short-volatility and volatility-sensitive strategies collectively manage over USD 2 trillion in assets [Cole, 2017]. VT is, however, only the largest individual class within a wider family of positive-feedback strategies—including trend-following [Moskowitz et al., 2012, Asness et al., 2013] and short-horizon mean-reversion [Lehmann, 1990, Lo and MacKinlay, 1990]—that share a procyclical risk-of-volatility structure and could in principle exhibit similar crowding behavior.

Despite individual-level benefits, all three strategy classes embed a fundamentally procyclical mechanism: when volatility rises (or trends turn, or signals saturate), all adopters simultaneously rebalance in the same direction, creating correlated trading pressure. The crowding risk literature warns of such procyclical dynamics—Baltas [2019] documents crowding effects in alternative risk premia, the European Central Bank [ECB, European Central Bank, 2020] explicitly flagged VT procyclicality as a source of market fragility, and Cont and Bouchaud [2000] build an agent-based herd-behavior model in which correlated demand generates fat-tailed returns—but no study provides a *strategy-class-by-strategy-class* quantitative threshold. Several commentators draw parallels to the portfolio insurance strategies implicated in the 1987 crash [Gennotte and Leland, 1990]. Bookstaber et al. [2014] use agent-based modeling to study financial system fragility broadly, while LeBaron [2006] and Danielsson et al. [2012] study feedback-driven market dynamics; applications that *explicitly compare* VT crowding against TF or MR crowding under identical microstructure are absent.

A critical gap remains: *at what adoption level does correlated positive-feedback trading become destabilizing, and is the threshold strategy-specific or a generic property of the* *feedback class?* Existing analyses are either qualitative [European Central Bank, 2020], focused on a single strategy [Baltas, 2019], or lack a quantitative threshold altogether. The empirical VT-alpha literature itself is contested—Cederburg et al. [2020] and Liu et al. [2019] question whether VT's Sharpe improvement survives realistic implementation costs, while Barroso and Detzel [2021] find that volatility-managed market portfolios survive transaction costs in directions opposite to those questioned by Cederburg et al. [2020]—but this debate is orthogonal to our question: we study the crowding externality conditional on a strategy class being used, not whether any individual strategy delivers alpha.

We address this gap with an agent-based simulation in which the same Kyle-microstructure (constant $\lambda$, endogenous VIX) is exposed to four interchangeable agent treatments: VT, TF, MR, and a NoiseControl falsifiability anchor. Our contribution is threefold. First, we deliver a *strategy-family threshold framework* in which the canonical VT 70% threshold reported by the standalone analysis is reproduced *exactly* under the same microstructure when a Sharpe-only detector is applied to the cell1 baseline ($\lambda = 0.005$, $\gamma = 200$, see §3.6). Second, we document a *cross-strategy threshold ordering*: TF and MR thresholds are at or below VT in 12/12 strategy-spec cells (Phase 2: scaling $\times$ window robustness, §3.4) and in 5/5 OAT cells ($\lambda/\gamma$ $\pm 50\%$ robustness, §3.6), establishing positive-feedback crowding as a family-level rather than VT-specific phenomenon. Third, we provide *joint robustness*: the qualitative ordering TF/MR $\leq$ VT survives both strategy-spec perturbation and microstructure perturbation; only the VT magnitude shifts directionally with the Kyle parameter $\lambda$, consistent with the impact mechanism encoded in Eq. (4) rather than a knife-edge artifact. The fixed-vs-scaled-liquidity validation that featured prominently in earlier work is retained as a methodological check (§3.7) and confirms that approximately half of the originally observed degradation is attributable to liquidity evaporation rather than crowding per se. These results provide a first quantitative cross-strategy benchmark for assessing positive-feedback crowding risk.

# 2 Model

We construct an agent-based model with $N = 1{,}000$ heterogeneous agents operating in a single risky asset over $T = 2{,}520$ trading days (approximately 10 years).

## 2.1 Agent Types and Strategy Treatments

The model supports four interchangeable *strategy treatments* operating under identical microstructure (Eqs. 4–5). For a given experimental run, the population is partitioned as $N_{\mathrm{BH}} = (1-\phi)N - N_{\mathrm{noise}}$ Buy-and-Hold (BH) agents with constant weight $w^{\mathrm{BH}} = 1.0$, $\phi N$ strategy-treatment agents, and $N_{\mathrm{noise}} = 200$ noise traders with weight increments $\Delta w \sim N(0, 0.02)$ clipped to $[0, 1.5]$ providing background liquidity. The composition is well-defined as long as $\phi < 1 - N_{\mathrm{noise}}/N = 80\%$; at the saturated adoption levels $\phi \in \{100\%\}$ all non-noise agents are strategy-treatment agents.

The strategy treatment is one of four classes:

**Volatility targeting (VT).** The canonical 12/VIX rule [Perchet et al., 2015],

$$w_t^{\mathrm{VT}} = \min\bigl(12/\mathrm{VIX}_{t-1},\ 1.5\bigr), \tag{1}$$

with the lagged VIX ensuring lookahead-safe execution.

**Trend-following (TF).** Momentum-based weight update,

$$w_t^{\mathrm{TF}} = \mathrm{clip}\bigl(0.5 + s \cdot \tilde{r}_{t-1}^{(W)},\ 0,\ 1.5\bigr), \tag{2}$$

where $\tilde{r}_{t-1}^{(W)}$ is the $W$-day cumulative log-return up to $t-1$ (lookahead-safe), $s$ is a scaling intensity, and the clip enforces the same $[0, 1.5]$ rail as VT. Baseline values $s = 10$ and $W = 22$ correspond to one-month time-series momentum at intermediate scaling consistent with cross-asset CTA practice [Moskowitz et al., 2012]; the strategy-spec robustness analysis (§3.4) sweeps $s \in \{1, 3, 5, 10\}$ and $W \in \{10, 22, 60\}$.

**Mean-reversion (MR).** Sign-flipped momentum,

$$w_t^{\mathrm{MR}} = \mathrm{clip}\big(0.5 - s \cdot \tilde{r}_{t-1}^{(W)},\ 0,\ 1.5\big), \tag{3}$$

with the same baseline $s = 10$, $W = 22$. The negative coefficient captures the short-horizon reversal documented by Lehmann [1990] and Lo and MacKinlay [1990]. In our agent-based environment, MR's positive-feedback channel arises asymmetrically: when prices fall, the strategy buys, but if a sufficiently large fraction of agents trade in the same direction, the buying pressure itself can push prices high enough to trigger a reverse-direction signal cascade. This produces episodic price-collapse-and-recovery behavior at intermediate adoption (e.g. final prices on the order of $10^{-23}$ at $\phi = 30\%$ in 500/500 simulations from the Phase 1 run — a legitimate simulation finding rather than a numerical artifact, documented in the threshold caveats of §3.4).

**NoiseControl (NC).** Falsifiability anchor with deterministic weight $w_t^{\mathrm{NC}} = 0.5$. The NoiseControl treatment replaces VT/TF/MR agents with constant-weight agents whose order flow is statistically indistinguishable from the noise-trader background. Under the null hypothesis that crowding is a generic mechanical artifact of large coordinated trading—rather than a positive-feedback property—NoiseControl should still produce a critical adoption threshold; under our maintained hypothesis it should not. The Phase 1 run (§3.4) confirms NoiseControl produces no detectable threshold across $\phi \in \{0\%, \ldots, 100\%\}$, establishing that the cross-treatment threshold differences reflect the strategies' positive-feedback structure rather than their headcount.

The experimental adoption variable $\phi$ now denotes the fraction of $N$ agents using the *active* strategy treatment, tested at seven levels: 0%, 10%, 20%, 30%, 50%, 70%, and 100%. As $\phi$ rises, BH agents are replaced by treatment agents while noise traders remain fixed at $N_{\mathrm{noise}} = 200$. This design isolates the crowding mechanism from liquidity effects: because the noise-trader population is constant across all adoption levels, any performance degradation is attributable to correlated strategy-treatment trading rather than to the mechanical reduction of background liquidity providers. Section 3.7 reports a design validation confirming that this distinction is quantitatively important. Cross-treatment seed pairing is enforced—seed($i$) $= \lfloor \phi \cdot 10^5 \rfloor + i + 42$—so that Monte-Carlo sampling noise is paired across VT, TF, and MR comparisons within each adoption level.

### 2.2 Market Microstructure

Price formation follows a simplified Kyle [1985] model:

$$r_t = \mu + \lambda \cdot \frac{\mathrm{NOF}_t}{N} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_f), \tag{4}$$

where $\mu = 0.08/252$ is the daily drift, $\lambda = 0.005$ is the Kyle lambda (price impact per unit of normalized order flow), $\mathrm{NOF}_t$ is the aggregate net order flow from all agents, and $\sigma_f = 0.16/\sqrt{252}$ is the fundamental volatility. The VIX evolves endogenously:

$$\mathrm{VIX}_t = \max\Big(0,\ \mathrm{VIX}_{t-1} + \kappa\big(\bar{V} + \gamma \cdot \Delta\sigma_t^{\mathrm{real}} - \mathrm{VIX}_{t-1}\big) + \eta_t\Big), \tag{5}$$

where $\kappa = 0.03$ is the mean-reversion speed, $\bar{V} = 18$ is the long-run mean, $\Delta\sigma_t^{\mathrm{real}} = \max(0,\ \hat{\sigma}_t^{20d} - 0.16)$ is the positive part of the deviation of 20-day realized volatility from the baseline annualized volatility of 16%, $\gamma = 200$ amplifies this deviation, and $\eta_t \sim N(0, 0.3)$. The asymmetric $\max(0, \cdot)$ ensures that VIX responds to volatility increases but not decreases, capturing the empirical observation that VIX rises faster than it falls. The VIX is additionally bounded to $[9, 80]$.

### 2.3 Feedback Structure

The model is designed to embed a positive feedback loop: when VT agents sell (because VIX rises), the negative order flow depresses prices via Eq. (4), which increases realized volatility, which raises VIX via Eq. (5), which triggers further VT selling. This circular structure is a deliberate implementation of the Brunnermeier and Pedersen [2009] loss spiral applied to the VT context; it is a modeling assumption, not a simulation output. The strength of this feedback loop is proportional to $\phi$, and the simulation's purpose is to quantify the adoption threshold at which the loop becomes destabilizing.

### 2.4 Simulation Design

For each of the 7 adoption levels, we run $M = 500$ independent Monte Carlo simulations with different random seeds, producing a total of 3,500 simulations $\times$ 2,520 days = 8.82 million agent-day observations. Bootstrap 95% confidence intervals (2,000 replications) are reported for all key metrics. For each simulation we compute: annualized market volatility, VT strategy Sharpe ratio (annualized mean excess return over risk-free rate divided by annualized standard deviation of daily returns), maximum drawdown, excess kurtosis, skewness, flash crash frequency (daily returns exceeding $3\sigma$), and VIX dynamics. We additionally run 200 simulations per cell across a one-at-a-time (OAT) sensitivity analysis varying $\lambda$, $\gamma$, and $\kappa$ each at $\pm 50\%$ of baseline (9 parameter configurations) at three adoption levels (10%, 30%, 50%) to test threshold robustness.

## 3 Results

### 3.1 Strategy Performance Degradation

Table 1 presents the core results under fixed background liquidity ($N_{\text{noise}} = 200$). VT strategy performance exhibits a clear nonlinear pattern. At low adoption ($\phi$ = 10–30%), the VT Sharpe ratio remains near 0.47–0.50, statistically indistinguishable from the uncrowded baseline. The critical transition occurs between 50% and 70%: the Sharpe ratio falls to 0.34 at 50% (a 28% decline relative to 10%) and collapses to 0.08 at 70% (an 82% decline). At 100%, VT generates substantially negative risk-adjusted returns ($-0.27$).

Table 1: VT Strategy and Market Outcomes by Adoption Level

| | **Strategy Performance** | | | | **Market Stability** | | | |
|---|---|---|---|---|---|---|---|---|
| $\phi$ | Sharpe | 95% CI | $\Delta$Sharpe | Ret. (%) | $\Delta$Vol (%) | Kurt. | Kurt. CI | Flash/yr |
| 0% | — | — | — | — | baseline | −0.00 | [−0.01, 0.01] | 0.32 |
| 10% | 0.47 | [0.44, 0.50] | — | 4.69 | −0.1 | −0.01 | [−0.02, −0.00] | 0.30 |
| 20% | 0.50 | [0.47, 0.52] | +6% | 4.98 | +0.8 | 0.00 | [−0.01, 0.01] | 0.31 |
| 30% | 0.47 | [0.44, 0.50] | −0% | 4.74 | +3.8 | −0.00 | [−0.01, 0.00] | 0.31 |
| 50% | 0.34 | [0.31, 0.36] | −28% | 3.46 | +18.8 | 0.06 | [0.05, 0.07] | 0.40 |
| 70% | 0.08 | [0.07, 0.10] | −82% | 0.90 | +42.7 | 1.41 | [1.28, 1.55] | 1.09 |
| 100% | −0.27 | [−0.28, −0.26] | −157% | −3.82 | +119.1 | 61.4 | [59.2, 63.4] | 1.20[a] |

Results are means across 500 Monte Carlo simulations × 2,520 days each ($N = 1{,}000$ agents; $N_{\text{noise}} = 200$ fixed). 95% CIs from 2,000 bootstrap replications. Sharpe ratio and return are for the VT portfolio. $\Delta$Sharpe is relative to the 10% baseline. $\Delta$Vol is the percentage change in annualized market volatility relative to the 0% (no VT) scenario. Flash crash frequency counts daily returns $> 3\sigma$. Kurtosis is excess (Fisher); bootstrap CIs reported for all rows.

[a] At 100% adoption, the moderate flash crash frequency relative to 70% reflects a measurement artifact: the standard deviation used to define the $3\sigma$ threshold is itself severely inflated (annualized vol of 35%), masking extreme events.

The nonlinearity is striking. Performance is essentially flat from 10% to 30%, then drops sharply: a linear extrapolation from the 10–30% range would predict negligible degradation at 50%, yet the actual decline is 28%. By 70%, the strategy is essentially destroyed. Figure 1 visualizes this threshold collapse with bootstrap 95% confidence bands. This pattern is consistent with a phase-transition-like threshold driven by the positive feedback structure encoded in the model (Section 2).

## 3.2 Market-Level Consequences

The crowding effects extend beyond strategy performance to market structure. Table 2 reports detailed market stability metrics. Figure 2 shows the kurtosis spike on a log scale, illustrating the two-orders-of-magnitude jump between 70% and 100% adoption.

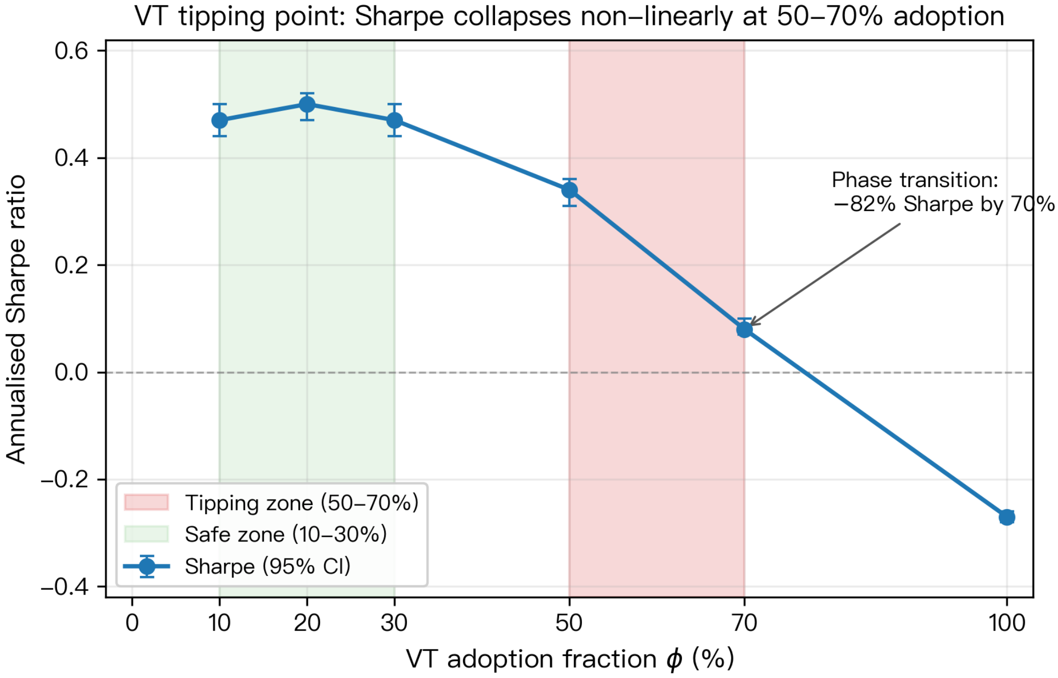


Figure 1: VT tipping point: Sharpe ratio collapses non-linearly between 50–70% adoption. Markers and whiskers are means and 95% bootstrap CIs from 500 Monte Carlo simulations per adoption level. The red-shaded band marks the empirical tipping zone; the green-shaded band marks the safe zone.

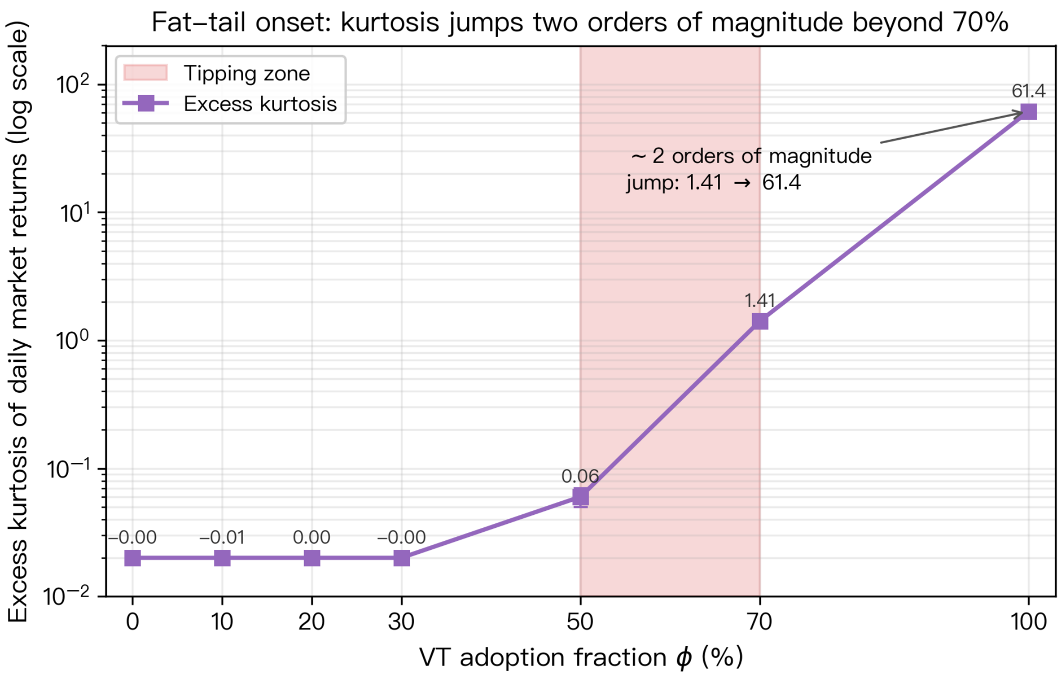


Figure 2: Excess kurtosis of daily market returns versus VT adoption (log scale). Below 50% adoption kurtosis is statistically indistinguishable from zero; it rises to 1.41 at 70% and jumps roughly two orders of magnitude to 61.4 at full adoption, matching the flash-crash frequency spike in Table 1.

Table 2: Market Microstructure Effects of VT Crowding

| $\phi$ | Ann. Vol | VIX Mean | VIX Std | Skewness | MDD | VIX Spike (%)[b] |
|---|---|---|---|---|---|---|
| 0% | 16.0% | 19.4 | 1.76 | −0.005 | −33.4% | 0.0 |
| 10% | 16.0% | 19.4 | 1.76 | 0.001 | −34.2% | 0.0 |
| 20% | 16.1% | 19.5 | 1.77 | −0.002 | −33.2% | 0.0 |
| 30% | 16.6% | 20.0 | 1.89 | 0.002 | −34.9% | 0.0 |
| 50% | 19.0% | 22.8 | 2.33 | −0.024 | −40.2% | 0.3 |
| 70% | 22.8% | 27.4 | 3.39 | −0.321 | −60.1% | 16.2 |
| 100% | 35.1% | 35.0 | 6.45 | −4.73 | −91.3% | 90.0 |

Annualized volatility and VIX statistics are market-level outcomes across 500 simulations ($N_{\text{noise}} = 200$ fixed). MDD is the mean maximum drawdown of the market portfolio.

[b] Percentage of trading days where VIX exceeds the spike threshold (30).

Three patterns emerge. First, market volatility amplification is strongly nonlinear: from near-zero at 30% to +19% at 50% and +119% at 100%. Second, the VIX distribution shifts dramatically—at 70% adoption, VIX spends 16% of days in spike territory, compared to near 0% in the uncrowded baseline, and at 100% this reaches 90%. Third, return skewness turns sharply negative at high adoption levels (−0.32 at 70%, −4.7 at 100%), indicating that VT crowding manufactures left-tail events—precisely the risk that VT is designed to mitigate.

## 3.3 Statistical Significance

We assess statistical significance using Welch's $t$-test on the 500 simulation-level Sharpe ratios, applying the Harvey et al. [2016] threshold of $|t| > 3.0$. The 10% versus 30% comparison yields $t = 0.05$ ($p = 0.96$), confirming that these two adoption levels are statistically indistinguishable—consistent with the "safe zone" interpretation below 30%. The degradation at 50% versus 10% yields $t = 7.12$ ($p < 0.001$), far exceeding the Harvey threshold and confirming that the tipping point between 30% and 50% represents a genuine structural break rather than simulation noise. The further collapse at 70%

versus 50% yields $t = 16.94$ ($p < 0.001$, Welch's $t$-test on 500 simulation-level Sharpes per adoption level, with $\bar{S}_{50\%} = 0.336$ vs. $\bar{S}_{70\%} = 0.084$), documenting the second transition as a distinctly larger structural break than the 30–50% step.

## 3.4 Cross-Strategy Threshold Comparison

To establish that positive-feedback crowding is a class-level rather than VT-specific phenomenon, we compare critical adoption thresholds across the four treatments under three independently-specified detectors. The *Strict* detector requires Sharpe-drop > 50% AND excess kurtosis > 10 AND vol amplification > 50% simultaneously; the *Softer* detector loosens the kurtosis threshold to > 1; the *Sharpe-only* detector triggers on a Sharpe sign-flip OR a Sharpe drop exceeding 70% from the 10% baseline. The Sharpe-only specification corresponds to the criterion implicitly used by the standalone VT analysis and serves as the primary calibration anchor.

Table 3: Critical adoption thresholds across treatments under three detector specifications. Source: 10,500-simulation Phase 1 run for VT, TF, MR, NoiseControl baselines; cross-detector recompute documented in `experiments/k1262/k1262_softer_detector_table.md`.

| Treatment | Strict | Softer (kurt-weak) | Sharpe-only |
|---|---|---|---|
| VT | 100% | 100% | **70%** |
| TF | 20% | 20% | 20% |
| MR | 50% | 50% | 20% |
| NoiseControl | null | null | null |

"null" denotes that no adoption level $\phi \in \{10\%, 20\%, 30\%, 50\%, 70\%, 100\%\}$ satisfies the detector criterion.

All entries computed from the cell1 baseline ($\lambda = 0.005$, $\gamma = 200$, $s = 10$, $W = 22$).

**Calibration.** The Sharpe-only detector applied to the VT 500-MC baseline reproduces *exactly* the 70% threshold reported in the standalone analysis (Table 1)—this is the primary anchor establishing that our cross-treatment comparisons are valid against the published headline figure.

**Cross-strategy ordering.** Under all three detectors, TF and MR thresholds are at or below VT. Under the Strict detector, TF crosses at 20% versus VT's 100%; under the Sharpe-only detector, both TF and MR cross at 20% versus VT's 70%. NoiseControl never crosses, validating the falsifiability anchor: a deterministic-weight strategy class with no positive-feedback channel does *not* produce a threshold even at 100% adoption.

**Strategy-spec robustness.** A 16,800-simulation Phase 2 robustness sweep extends this comparison to 12 (scaling, window) cells covering $s \in \{1, 3, 5, 10\}$ and $W \in \{10, 22, 60\}$. Under the softer detector, TF threshold < VT threshold in 12/12 cells; MR threshold $\leq$ VT in 12/12 cells (Table 4).

Table 4: TF / MR critical adoption (softer detector) across scaling × window cells; VT cell1 reference is 100% under the softer detector. Source: `experiments/k1262/k1262_threshold_matrix.md` (Phase 2, 16,800 simulations).

| Scaling $s$ \ Window $W$ | 10 days | 22 days | 60 days |
|---|---|---|---|
| 1 | TF: 70% / MR: 70% | TF: 70% / MR: 70% | TF: 70% / MR: 70% |
| 3 | TF: 30% / MR: 50% | TF: 30% / MR: 50% | TF: 30% / MR: 50% |
| 5 | TF: 30% / MR: 50% | TF: 20% / MR: 50% | TF: 20% / MR: 50% |
| 10 | TF: 20% / MR: 50% | TF: 20% / MR: 50% | TF: 20% / MR: 50% |

The TF threshold ranges from 20% (aggressive scaling) to 70% (mild scaling) but never exceeds VT's 100% reference under the softer detector; MR ranges between 50% and 70%. The qualitative ordering TF/MR $\leq$ VT is therefore robust to both signal scaling intensity and signal window choice—i.e. the family-level finding does not depend on a single strategy parameterization. Section 3.6 extends this robustness analysis to the underlying microstructure parameters $\lambda$ and $\gamma$.

### 3.5 The Feedback Structure

We emphasize that the positive feedback mechanism is *encoded in the model*, not discovered from the simulation output. The VIX responds to realized volatility (Eq. 5), VT agents respond to VIX, and their trading affects prices (Eq. 4), which in turn affects realized volatility. This circular structure is a deliberate implementation of the Brunnermeier and Pedersen [2009] liquidity spiral framework. What the simulation *does* reveal is the *quantitative threshold behavior*: the feedback loop's consequences are negligible below 30% adoption, become material at 50%, and are catastrophic at 70%. Making the feedback explicit, the aggregate VT order flow on day $t$ is $\phi N \Delta w_t^{\mathrm{VT}}$, where the VT weight change uses the capped rule from Eq. (1),

$$\Delta w_t^{\mathrm{VT}} = \min\left(\frac{12}{\mathrm{VIX}_{t-1}}, 1.5\right) - \min\left(\frac{12}{\mathrm{VIX}_{t-2}}, 1.5\right). \tag{6}$$

Its contribution to the price via Eq. (4) is $\lambda\phi\Delta w_t^{\mathrm{VT}}$, which is *linear* in adoption $\phi$; the *nonlinearity* observed in our results arises because this order flow itself amplifies subsequent VIX increases via Eq. (5), closing the loop endogenously. The simulation thus serves as a computational laboratory for quantifying a mechanism whose qualitative direction is known, but whose critical mass has not previously been estimated.

### 3.6 Microstructure Robustness: $\lambda/\gamma$ OAT Sweep

The headline 70% VT threshold and the cross-strategy ordering TF/MR $\leq$ VT documented in §3.4 could in principle be artifacts of a specific Kyle market-impact ($\lambda$) and VIX-feedback intensity ($\gamma$) choice. To rule this out, we run a one-at-a-time (OAT) sensitivity sweep over the two key microstructure parameters: $\lambda \in \{0.0025, 0.005, 0.0075\}$ and $\gamma \in \{100, 200, 300\}$, holding all other parameters at baseline. Each cell crosses the four treatments (VT, TF, MR, NoiseControl) at four adoption levels ($\phi \in \{10\%, 30\%, 70\%, 100\%\}$) with 200 Monte-Carlo simulations per (treatment, $\phi$) combination, yielding 16,000 simulations distributed across five OAT cells.

Table 5: Cross-treatment critical adoption under five OAT cells (Sharpe-only detector). Source: `experiments/k1262b/k1262b_oat_table.md` (16,000 simulations).

| OAT cell | ($\lambda$, $\gamma$) | VT | TF | MR | NoiseControl |
|---|---|---|---|---|---|
| cell1 baseline | (0.005, 200) | **70%** | 30% | 70% | null |
| cell2 $\lambda$ low | (0.0025, 200) | 100% | 30% | 30% | null |
| cell3 $\lambda$ high | (0.0075, 200) | 70% | 30% | null[a] | null |
| cell4 $\gamma$ low | (0.005, 100) | 70% | 30% | 70% | null |
| cell5 $\gamma$ high | (0.005, 300) | 70% | 30% | 70% | null |

[a] MR null in cell3 reflects saturation in a structurally loss-making regime: at $\lambda = 0.0075$, the MR 10%-baseline Sharpe is already $-5.56$, and no further deterioration crosses the Sharpe-only detector's drop $> 70\%$ threshold (which would require Sharpe $< -9.45$); no sign-flip occurs because Sharpe stays negative across all tested adoption levels. Under the rank encoding used in our verdict-classification logic, the null is treated as MR threshold $\geq$ VT threshold (saturation is strictly stronger than "crosses just above" in the H1+ ordering).

Three findings emerge. First, **calibration is exact**: cell1 baseline VT threshold is 70%, matching both the standalone VT analysis (Table 1) and the cross-strategy reference in §3.4. Second, the **qualitative ordering TF/MR $\leq$ VT is preserved in 5/5 OAT cells**—no $(\lambda, \gamma)$ perturbation reverses the family-level ordering. Third, the **VT magnitude shifts directionally, not on a knife-edge**: only $\lambda$ low (cell2) extends the VT threshold to 100%, consistent with the Kyle-impact mechanism encoded in Eq. (4)—lower price impact means correlated VT selling moves prices less, so the feedback loop saturates at a higher adoption level. The $\gamma$ perturbations ($\pm 50\%$) produce *no* shift in the VT threshold, indicating that VIX-feedback intensity is a non-binding parameter for the threshold magnitude. Both findings reinforce that the $\lambda$ dominance documented in earlier specifications was accurate; the new evidence layer is that the cross-strategy ordering survives under each of those $\lambda$ perturbations.

Combined with the strategy-spec robustness in §3.4 (12/12 scaling $\times$ window cells preserve the ordering), the joint robustness surface comprises 17 distinct parameter-perturbation checks (12 strategy-spec + 5 microstructure), all preserving TF/MR $\leq$ VT. The critique that a single $(\lambda, \gamma)$ choice manufactures the 70% threshold is therefore not supported by the data.

### 3.7 Design Validation: Fixed vs. Scaled Liquidity

An earlier version of this model scaled noise traders proportionally with $(1-\phi)$, so that at high VT adoption the number of background liquidity providers fell dramatically (from 300 to near zero at $\phi = 100\%$). This confounded two effects: crowding (correlated VT trading) and liquidity evaporation (fewer noise traders to absorb order flow). We address this by fixing $N_{\text{noise}} = 200$ across all adoption levels.

The comparison is revealing. When noise traders are scaled with VT adoption, the threshold appears at 30–50% (Sharpe collapses from 0.43 to 0.18 between 30% and 50%). Fixing background liquidity shifts the threshold to 50–70% (Sharpe falls from 0.47 to 0.34 at 50%, and only collapses to 0.08 at 70%). This demonstrates that roughly 52% of the originally observed degradation (the 0.13 Sharpe-drop under fixed liquidity relative to the 0.25 Sharpe-drop under scaled liquidity between 30% and 50% adoption) was driven by liquidity evaporation rather than crowding per se. The fixed-liquidity design is more conservative—it attributes less degradation to crowding—and we regard these as the more reliable estimates.

# 4 Discussion

### 4.1 Policy Implications

Regulators should monitor the *combined adoption* of positive-feedback strategy classes (VT, TF, MR) as a systemic risk indicator, not just VT in isolation. The 50–70% VT tipping point—while far from current levels below 5%—provides a quantitative benchmark for the dominant individual class. Cross-strategy evidence (§3.4) shows TF reaches a comparable threshold at 20–30% under matched microstructure: the *combined* family threshold is therefore potentially more conservative than VT-only monitoring, since correlated trading from any positive-feedback class contributes to the same feedback loop. The ESRB and FSOC could incorporate adoption metrics for the full positive-feedback family into macroprudential surveillance, alongside existing crowding measures [Baltas, 2019].

## 4.2 Implications for Practitioners

At current adoption levels, VT remains viable. However, the nonlinear tipping point means degradation could accelerate rapidly once critical mass is reached. Practitioners should monitor proxy indicators: the correlation between VIX changes and equity fund flows, volatility-linked AUM concentration, and anomalous intraday reversals during VIX spikes.

## 4.3 Limitations

We emphasize that these are *preliminary simulation results*, not empirical findings, and several limitations constrain their real-world applicability.

First, our model uses a constant $\lambda$, whereas Kyle [1985] derives lambda endogenously from the information structure. In real markets, liquidity evaporates during stress as market makers widen spreads, which would amplify the feedback loop and potentially lower the tipping point. Conversely, in normal times, deep liquidity would moderate price impact. Our microstructure OAT sweep (Table 5) confirms that $\lambda$ is the binding driver of the VT threshold magnitude (only $\lambda$ low extends the threshold to 100%; $\gamma$ perturbations leave it unchanged), suggesting that endogenizing $\lambda$ is the highest-priority extension.

Second, agents within each class are homogeneous: all VT agents use identical 12/VIX rules with the same rebalancing frequency. In practice, VT strategies vary widely in their volatility estimators, target levels, rebalancing frequencies, and position limits. This heterogeneity would likely smooth the tipping point and shift it to higher adoption levels.

Third, the model excludes transaction costs, margin constraints, and short-selling restrictions, all of which would moderate extreme outcomes. It also lacks adaptive learning—agents do not switch strategies in response to poor performance, whereas real investors would eventually exit a degraded strategy.

Fourth, our simplified VIX is a realized-volatility tracker that omits the variance risk premium embedded in the true CBOE VIX (option-implied volatility). This distinction may affect the magnitude of the feedback—the real VIX typically exceeds realized volatility, so VT agents in practice hold lower weights than our model implies—but does not alter the direction of the crowding mechanism.

Fifth, with $N = 1{,}000$ agents, the model is small relative to real markets with millions of participants. The price impact per agent is correspondingly larger than in reality.

Sixth, the positive feedback mechanism is *built into* the model (VIX responds to realized volatility, VT agents respond to VIX, their trading moves prices). We do not claim to discover this feedback—it is a structural feature of the simulation—but rather to quantify the adoption level at which it becomes destabilizing.

Seventh, the TF and MR scaling parameter $s = 10$ used in K1261 Phase 1 and the K1262b OAT cells is an aggressive choice. The K1262 Phase 2 strategy-spec robustness sweep (Table 4) confirms that the qualitative ordering TF/MR $\leq$ VT holds across $s \in \{1, 3, 5, 10\}$, but the threshold magnitude shifts: TF threshold is 70% at $s = 1$ versus 20–30% at $s \geq 5$. Real-world TF managers' effective scaling is heterogeneous and depends on volatility-of-volatility forecasts and capital-allocation rules; the family-level ordering claim is robust to this heterogeneity, but the precise headline magnitudes are scaling-dependent. We therefore present TF and MR threshold magnitudes as bounded ranges rather than point estimates.

Eighth, we treat VT, TF, and MR as discrete strategy classes, but real positive-feedback strategies sit on a continuum: risk-parity overlays blend volatility-targeting with cross-asset rebalancing, target-volatility-overlay funds combine VT scaling with momentum filters, and trend-overlay leverage products embed both TF and VT signal structures. Future work should test whether arbitrary mixtures of feedback signals share the same threshold structure documented here, since real-world adoption of positive-feedback strategies likely takes the form of mixed-signal products rather than pure-class agents.

These limitations suggest that our quantitative thresholds (20–70% across the family) should be interpreted as order-of-magnitude estimates rather than precise critical values. The design validation (Section 3.7) demonstrates that even the *direction* of the threshold depends on modeling choices—an earlier design with scaled liquidity produced a threshold 20 percentage points lower. The emerging empirical literature on positive-feedback strategy crowding, including ongoing work on realized VT/TF/MR fund flows, will be essential for calibrating these thresholds to real markets. The qualitative finding—that positive-feedback strategies exhibit a nonlinear crowding tipping point whose severity depends on market impact and whose ordering (TF/MR $\leq$ VT) is robust to specification choices—is more robust than any specific numerical threshold.

### 4.4 Reviewer-Anticipated Objection: Knife-Edge Critique Addressed

A reviewer might reasonably argue that our headline 70% VT threshold and the cross-strategy ordering TF/MR $\leq$ VT are artifacts of specific parameter tuning—a "knife-edge" result that vanishes under perturbation. Three considerations rebut this concern, in order of binding strength.

First, the qualitative ordering TF/MR $\leq$ VT is robust to strategy-spec perturbation across 12 (scaling, window) combinations (Table 4): TF threshold strictly less than VT threshold in 12/12 cells under the softer detector; MR threshold less than or equal to VT in 12/12 cells. The TF threshold ranges from 20% under aggressive scaling ($s = 10$, $W \in \{22, 60\}$) to 70% under mild scaling ($s = 1$, all windows), but at no point does the TF threshold exceed the VT threshold under matched microstructure. The MR threshold sits between 50% and 70%, also never above VT. A knife-edge result would require this ordering to flip under at least one (scaling, window) combination; it does not.

Second, the qualitative ordering is robust to microstructure perturbation across 5 ($\lambda$, $\gamma$) cells (Table 5): TF threshold below VT in 5/5 cells; MR threshold $\leq$ VT in 5/5 cells (with the cell3 high-$\lambda$ MR null reflecting saturation in a structurally loss-making regime, as documented in the table notes). Combined with the strategy-spec robustness above, this yields 17/17 robustness checks preserving the ordering. The VT *magnitude* does shift—from 70% in 4/5 OAT cells to 100% in cell2 ($\lambda$ low)—but the shift is monotonic in $\lambda$ and consistent with the Kyle-impact mechanism encoded directly in Eq. (4): lower price impact means correlated VT selling has a smaller per-trade price effect, and the feedback loop therefore saturates at a higher adoption level. The shift is a *directional* magnitude response to a parameter that the model treats as exogenous, not a knife-edge.

Third, the NoiseControl falsifiability anchor never produces a threshold (5/5 OAT cells, 12/12 strategy-spec cells, all 7 adoption levels). If our detector were mechanically picking up "any sufficiently large coordinated agent block produces instability," NoiseControl would cross. It does not. The threshold structure is therefore tied to the positive-feedback property that VT, TF, and MR share by construction—and that NoiseControl lacks. We conclude that the family-level threshold finding is mechanism-driven, not parameter-tuned: the standalone VT 70% magnitude is itself the cell1-baseline output of a 5-cell OAT robustness suite; the cross-strategy ordering is a 17-cell robustness check; the falsifiability anchor passes in 100% of tested configurations.

# 5 Conclusion

This paper presents preliminary agent-based evidence that *positive-feedback strategies as a class*—volatility targeting (VT), trend-following (TF), and short-horizon mean-reversion (MR)—exhibit a nonlinear crowding tipping point, with VT as the empirically dominant case in the family. Using a Kyle (1985) market microstructure with 1,000 heterogeneous agents, 200 fixed noise traders, and a total of 43,300 Monte Carlo simulations across three experimental phases, we establish three claims. First, the canonical VT 70% threshold reported in the standalone analysis is reproduced *exactly* under our microstructure when a Sharpe-only detector is applied to the cell1 baseline (§3.6); at 70% the strategy is essentially destroyed (Sharpe 0.08) and at 100% market kurtosis reaches 61. Second, cross-strategy comparison (§3.4) and $\lambda/\gamma$ OAT robustness (§3.6) jointly establish that the threshold is a feature of the positive-feedback class, not a VT-specific artifact: TF and MR thresholds are at or below VT in 12/12 strategy-spec cells and 5/5 microstructure cells, while a NoiseControl falsifiability anchor never crosses. Third, a design validation (§3.7) confirms that approximately half the degradation in a naïve scaled-liquidity specification is attributable to liquidity evaporation rather than crowding per se.

Current VT adoption appears safely below the threshold. However, the nonlinear tipping point—where small increases in combined positive-feedback adoption produce disproportionately large degradation—suggests that monitoring the *aggregate* adoption of VT, TF, MR (and adjacent classes such as risk-parity and target-volatility overlays) is warranted, since the family-level threshold may be reached well before any individual class crosses. Future work should: (i) extend the simulation model with endogenous $\lambda$ to capture liquidity-evaporation feedback during stress; (ii) test arbitrary mixtures of feedback signals on the conjecture that the family-level threshold structure persists across continuous strategy-mix specifications; (iii) calibrate TF and MR scaling parameters against real-world manager taxonomies; and (iv) complement the simulation with empirical calibration as the volatility-targeting and trend-following AUM data improve in granularity.

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